EMILY STEINER

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Education

Stanford University

MS-PhD Candidate In Electrical Engineering — ***2023-2028 (expected)***

- **Advisor**: Professor Iro Armeni, **Focus**: 3D/4D Computer Vision
- **Course Highlights:** Foundation Models for 3D/4D Scene Understanding, Deep Learning for Computer Vision, Computational Imaging

University of Waterloo

BASc Honours Mechatronics Engineering/Computing Option - Co-operative Program - With Distinction — ***2018 - 2023***

- Graduated Dean's Honour List - *Cumulative average 95.6%*
- **Course Highlights:** Foundations of AI, Pattern Recognition, Image Processing, Autonomous Mobile Robots, Automatic Control Systems

Research Experience

Gradient Spaces Lab | Graduate Research Assistant | *Advisor: Iro Armeni, Stanford University* — ***April 2024 - present***

- Architected and trained a query and transformer-based model for dynamic 3D (4D) scene understanding, implementing novel methods for sharing information across temporally sparse 3D observations to achieve temporally consistent semantic instance segmentation.
- Investigating language grounding and 4D semantic representations for applications in embodied AI for evolving living spaces and creating digital twins in construction, aimed at enhancing understanding of human usage patterns, documentation, and resource efficiency.

Computational Imaging Lab | Rotation PhD Student | *Advisor: Gordon Wetzstein, Stanford University* — ***Winter 2024***

Publications

ReScene4D: Temporally Consistent Semantic Instance Segmentation of Evolving Indoor 3D Scenes

Emily Steiner, Jianhao Zheng, Henry Howard-Jenkins, Chris Xie, Iro Armeni

Under Review, *2026*

Industry Experience

Lumafield | Hardware Research & Development Engineering Co-op — ***May - Aug 2022***

- Investigated CT scanner imaging limitations, identifying the balance between sharpness and scan time. Achieved a 3x speed increase with no quality loss, implementing changes on customer machines and presenting results at trade shows.
- Developed an exploratory multi-detector prototype and an image processing pipeline using OpenCV in Python. Implemented a joint iterative calibration algorithm that improved capture speed by 2x with minimal quality impact.

Inertia Product Development | Product Development Intern (Electrical / Mechatronics) — ***Jan - Apr 2022***

- Designed a LiDAR point cloud visualizer using Python and Qt to interface with a Robot Operating System (ROS) backend.
- Performed a feasibility study and utilized rapid prototyping to identify the best technical direction, contributing to project success.

Canadensys | Aerospace Engineer — ***May - Aug 2021***

- Engineered embedded state machine architecture and PID controller firmware for a high-precision BLDC motor using an STM32 microcontroller in C; performed control system and stability analysis with Simulink.
- Conducted root cause analysis to identify a hardware design flaw affecting feedback stability. Developed a corrective algorithm for Hall Effect sensor processing, successfully salvaging the PCB design.

ExactEarth (Spire) | Software Engineering Co-op Student — ***Jan - Apr & Sep - Dec 2020***

- Developed autonomous operations management software for the EV10 satellite, facilitating communication procedures, telemetry collection, and recovery processes. Supported software control during satellite commissioning (launched Sept 2020).

Leadership & Involvement

Teaching Assistant: Computer Vision for the Built Environment ↗, Stanford University — ***Winter 2026***

Stanford Women In Electrical Engineering (WEE) — ***2023-present***

- **Faculty Liaison** (2024-26): responsible for organizing quarterly faculty roundtables.
- **Co-Mentorship Chair** (2025-26): co-organizer and graduate mentor (2024-present) in the mentorship program and volunteer at STEM education outreach events (Stanford SPLASH, SLAC community day).

ICRA 2025 Challenge Organizer: Nothing Stands Still Challenge Hosted by Hilti ↗ — ***2025***

Undergraduate Mentor: University of Waterloo Mechatronics Mentorship Program — ***2020-2022***

Honors & Awards

2025	**TomKat Graduate Fellow in Translational Research ↗**, Stanford University
2022	**Co-Operative Education Student of the Year Honourable Mention ↗**, University of Waterloo
2022,21, 19	**First In Class Engineering Scholarship**, University of Waterloo
2019	**President's International Experience Award**, University of Waterloo
2018	**President's Scholarship of Distinction**, University of Waterloo

Projects

Probing the Object Awareness of Current 3D LLMs *Fall 2024*

- Designed and conducted systematic perturbation experiments revealing density-induced hallucination and bottlenecks in object detection capabilities of state-of-the-art 3D-LLM architectures.

Towards Open Scene Understanding For Construction Analysis *Spring 2024*

- Developed an integrated pipeline to lift 2D semantics from foundational vision models (e.g., GroundingDINO, SAM, CLIP) into unified 3D mesh representations, evaluating generalizability and robustness for zero-shot, query-driven monitoring on dynamic construction datasets, and demonstrating both the potential and current limitations of this approach.

Viability of Eye-Tracking Glasses for Beamforming Hearing-Aids *Winter 2024*

- Engineered and simulated a spatial audio steering system using microphone-enabled glasses, demonstrating enhanced source isolation via eye-tracking-driven beamforming for assistive hearing technology.

Drag Reduction System Automation | Capstone Project in collaboration with Williams Racing Formula 1 *Sept 2022 - Apr 2023*

- *1st Place Winner of the University of Waterloo Design Analysis Competition Sponsored by Ansys*
- Designed a system to automate movement between testing positions for the wind tunnel model's drag reduction system without affecting aerodynamic surfaces.
- Performed mathematical modelling of the airfoil surface during wind tunnel testing using Python (NumPy, scikit-learn, and SciPy) and MATLAB.
- Analyzed system performance and validated the design in adapted conditions to overcome testing limitations.

Comparison of ML Techniques for Freezing of Gait (FoG) Detection *Jan - Apr 2023*

- Investigated the trade-offs of various published machine learning (ML) techniques for the detection of FoG using tri-axial IMU sensors.
- Adapted pre-processing and data splitting techniques to consider performance for implementation on portable detection devices.

Technical Skills

Technical Skills Python (PyTorch & Lightning), 3D Vision (Pointcept, MinkowskiEngine, Open3D), Viz (Blender), C++ (familiarity)

Research Infrastructure Distributed Training (DDP), Large-scale Experiments (HPC clusters, W&B), Linux/Unix, Git, Docker